
A FORMAL PROOF OF THE TRISDUCTION COMPLETION IN PURE ALGEBRA

*From the Root Axiom's Kinetic Floor
through the Order-Sensitivity Completion
to the Twelve-Gate Torsor and the
Closed-Form Verdict*

Sealed · Substrate-Portable

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ABSTRACT

The Trisduction architecture carries three seals on one verification core: linguistic-semantic, topological-geometric, and mathematical. Anchor-level independence is the architecture's own law, and a seal that cannot stand with the other two instrument sets removed is not an independent seal. The standing presentation distributes load: the triaxial count is forced operationally by the deletion test, the twelve-gate cardinality is witnessed geometrically by Euler closure and Newton-Gregory packing, and the plurality of verification axes enters the algebraic completion as an explicit premise. This paper closes the question by rederiving the whole of the master codex's Part II (Islam 2026b) in pure algebra, with the linguistic and geometric instrument sets removed entirely and demoted to corroboration. The chain runs as follows. The Root Axiom is stated at axiom grade with its kinetic floor carried at theorem grade by the quantum speed limit. The composition law of the architecture is premise-typed to its legislative sources. An Order-Sensitivity Completion theorem then shows that non-commutativity of audit composition alone forces the quaternions $\mathbb{H}$ by the Frobenius classification, fixing exactly three orthogonal axes over one scalar ground and demoting the plurality floor from premise to corollary. The twelve-gate roster is recovered three times inside the integers of $\mathbb{H}$: by slot combinatorics, by the A_4 torsor with its binary tetrahedral double cover, and by the pure-imaginary slice of the norm-2 Hurwitz shell. The truth function stands in closed form, $\det(R) = (\text{Re}(\hat{q}_F \hat{q}_E \hat{q}_{ER}))^2$, with derived bounds, a Weyl-closed invariant catalog, conjugation frame-invariance, explicit precedence, and the Return Law $\text{GOL} \iff \lambda \neq 0$. The falsifiable core is an executable instrument and a sixteen-check recorded battery: identity residues at machine precision and parameter-free integer counts, twenty-four, twelve and twelve, and the class equation 1, 3, 4, 4. Any re-execution failure falsifies the corresponding claim. One premise, order-sensitivity, carries both the triaxial count and the directed roster. Mathematics alone seals the architecture it was already inside.

1. BACKGROUND AND RATIONALE • THE BARRIER

The architecture under audit carries its seals under an explicit independence law. Seal L, the linguistic-semantic seal, stands on the deletion-test discipline and the Linguistic Isolation Test with no geometry and no algebra anywhere in its anchors. Seal G, the topological-geometric seal, stands on the Root Axiom decomposition, Euler closure, Newton-Gregory packing, and the Operational Content Theorem with no multiplication anywhere in its anchors. Seal M, the mathematical seal, must stand on the architecture's composition law and the classification theorems of the real division algebras with no topology and no linguistics anywhere in its anchors. The law sets a test that can be failed: remove the other two instrument sets and watch whether the verdicts move. This paper executes that removal in full and reports the result.

The linguistic seal forces the triaxial count operationally. The Root Axiom parses into exactly three irreducible components under the deletion test with vocabulary disjointness, and the seal's own warrant typing is explicit on what this is and is not: the forcing is operational-procedural, reproducible across analysts applying the same discipline, and it is not a predicate-logic uniqueness theorem, with none claimed. Remove the linguistic instrument set and the count of three must arrive from somewhere else, or the architecture's central cardinality reduces to a convention.

The geometric seal closes the roster topologically. The four-vertex closure is certified by Euler's identity, and the twelve-fold cardinality of the directed gate roster is confirmed by the kissing number of three-dimensional space. Both witnesses are theorem-grade in their own registers, and both are geometric. Remove the geometric instrument set and the twelve must be recovered inside algebra alone, or the roster's cardinality stands on borrowed ground.

The third pressure point is subtler and sits inside the algebraic completion itself. The master codex completes the verification algebra to the quaternions under three composition clauses and a floor: associativity, integrality, linearity with ground identity, and Floor 1, axis plurality, the demand that at least two linearly independent verification axes exist. The clauses are premise-typed to the architecture's own legislation. The floor, as carried, is an additional premise. A premise that exists only to feed a classification theorem is a structural debt: the classification's own dichotomies should be able to repay it. Whether they can is a precise mathematical question, and this paper answers it affirmatively.

The wall that makes the question nontrivial is the closure failure of three-dimensional space. A verification structure with three axes and no scalar slot would live on $\mathbb{R}^3$, and $\mathbb{R}^3$ admits no closed associative composition without zero divisors. The obstruction is visible in bare coordinates: closing $\{1, i, j\}$ under an

associative product forces a structure constant satisfying $c^2 = -1$, an equation with no real solution, the exact equation that stopped Hamilton's triplets for thirteen years. The barrier is structural, not computational. No refinement of calculation closes $\mathbb{R}^3$, because the failure is an algebraic impossibility, not a numerical difficulty. Resolution requires a different carrier, and the inventory of admissible carriers is a closed classification.

Two bodies of mathematics sat unexploited on either side of that wall for roughly ninety years. The determinant of the correlation matrix has been a named statistical object since Wilks introduced the generalized variance in 1932, its vanishing the textbook multicollinearity diagnostic, the volume of evidential independence used negatively as a pathology detector and never positively as a truth functional with a derivation. The scalar triple product, the determinant, the signed volume, predates Hamilton by over a century in the work of Leibniz, Cramer, and Cauchy; what Hamilton's quaternion calculus of 1844 supplied is its closed algebraic carrier, the triple product appearing as the scalar part of a vector product, from which the dot and cross products were later extracted as fragments. The master codex reconnects them at the geometric register. The present register is stricter: the reconnection must stand with the geometric scaffolding removed, as a closed-form identity whose proof touches nothing but algebra.

A third fragment sits in the empirical literature. Quantum cognition has produced parameter-free evidence that human evidence-taking composes non-commutatively: the question-order effects of Wang and colleagues satisfy the QQ equality, a signature of non-commuting operations with no fitted constants. That literature operates in $\mathbb{C}$, the algebra carrying exactly one imaginary axis, which is the collinear degeneracy standing as an algebra. A single axis cannot host orthogonal warrant separation. The quaternionic register, three axes over one scalar, stood unoccupied. The codex occupies it; this paper shows the occupation is forced.

A fourth barrier is epistemic and binds the paper's own conduct. The Root Axiom is an axiom. An axiom is not a theorem, and no amount of derivation downstream converts it into one. The anchor-inflation failure mode, claiming theorem grade where the warrant is conditional or premised, is named in the architecture's own failure taxonomy and is forbidden by its warrant phrasing law. What mathematics does supply for the Root Axiom is exact and bounded: a transition-register floor theorem, the quantum speed limit, binding every actuation, reversible or not, conditional on the postulates of quantum dynamics. This paper states the axiom at axiom grade, states the floor at conditional theorem grade, and claims nothing further.

A fifth barrier is the gate-content fence, and it is load-bearing rather than embarrassing. The architecture's own sealed discipline carries the clause that gate content is never derived from the algebra: which pathology each directed

transition prevents is forced by the Operational Content Theorem on source-role and target-role pairings, an operational result, not an algebraic one. An algebra that claimed to generate operational content would break the architecture's law while pretending to complete it. The mathematics-alone register therefore has a precisely mapped validity domain: it delivers the carrier, the count, the direction law, the group, the shell, the closed-form truth functional, the invariant catalog, the frame invariance, and the precedence. It does not deliver the truth of the Root Axiom, the content of the gates, or the engineering of the quantization mapping, and it says so.

That domain map is the autopsy this register requires. The standing distribution of load across three instrument sets did not fail from lack of effort. It worked, and it sealed. What it left open is the counterfactual that the independence law itself demands: subtract two of the three instrument sets and verify the survivor closes alone. The subtraction is not a stylistic exercise. Substrate portability, the architecture's operational principle that any sufficiently capable substrate reproduces the verdicts from the statements as printed, is strongest exactly at the algebraic register, where convergence operates on discrete verdicts and parameter-free integer counts compared field by field, never on prose against prose and never on scalar entries.

The result, stated before the derivation: the barrier yields. Under the architecture's own composition law, the single premise that audit composition is order-sensitive forces the quaternions, and the quaternions carry everything else: the three axes as the negative eigenspace of the unique ground-fixing involution, the scalar slot as the center, the twelve as ordered-pair combinatorics quadruply rooted in the integers of $\mathbb{H}$, and the verdict as the squared scalar part of a composed triad. The plurality floor stops being a premise and becomes a corollary. The premise ledger of the architecture shrinks by one at this register, and the shrink is itself a finding.

The remainder of the paper executes this in full: the literature fragments and their common structural error in Section 2, the protocol and warrant discipline in Section 3, the derivation chain in Section 4, the executable instrument with its recorded battery and falsifier clauses in Section 5, the discussion of what the third seal closes in Section 6, and the self-contained Frobenius proof in Appendix A so that the paper carries its own classification engine, depending on no external text outside the three consumptions named in Section 3.

2. LITERATURE · THE UNASSEMBLED FRAGMENTS

Five bodies of work supply pieces of the present result. Each is summarized with structural precision, each is then located at the exact point where it stops.

The classification line. Frobenius proved in 1878 that the finite-dimensional associative real division algebras are exactly $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{H}$. Hurwitz proved in 1898 that a multiplicative quadratic norm composes only in dimensions 1, 2, 4, and 8, and Zorn's 1933 analysis placed the octonions as the last alternative division algebra. The topological completions of Bott and Milnor, Kervaire, and Adams later closed every non-associative escape route, confining real division structure of any kind to dimensions 1, 2, 4, 8. The line delivers a complete and closed inventory of admissible carriers. What it does not deliver is any bridge from verification practice to its own hypotheses. Associativity, absence of zero divisors, and linearity are inputs to the classification, and a verification architecture that merely assumes them has assumed its way to the answer. The hypotheses must be earned from the architecture's own legislation, clause by clause, or the classification fires on nothing.

The generalized-variance line. Wilks named $\det(R)$, the determinant of the correlation matrix, as the generalized variance in 1932, and Bartlett supplied its null-distribution machinery in 1951. The object has lived in the statistics literature for nearly a century as a diagnostic: its vanishing detects multicollinearity, its smallness warns of redundancy. The line delivers the verdict object with a developed sampling theory. What it does not deliver is a positive derivation. In this literature $\det(R)$ is a pathology detector consulted after the fact, never a truth functional derived from first structure, and nothing in the statistical tradition explains why this determinant, rather than any other functional of the evidence, should carry a verdict.

The invariant-theoretic line. Weyl's first fundamental theorem for the rotation group generates all polynomial invariants of vector tuples in $\mathbb{R}^3$ from pairwise inner products and 3×3 determinants. The line delivers catalog-closure machinery of complete generality. What it does not deliver is selection: among the closed catalog of invariants, which one is the verdict, why it appears squared, and what algebra the square is the shadow of are questions invariant theory is not built to answer.

The quantum-cognition line. Busemeyer and Bruza assembled the case that human judgment composes by non-commuting operations, and Wang, Solloway, Shiffrin, and Busemeyer confirmed the QQ equality on question-order data with no fitted parameters, a rare parameter-free empirical signature. The line delivers exactly the empirical face of the premise this paper runs on: evidence-taking is order-sensitive. What it does not deliver is the carrier. The formalism operates in

complex Hilbert space, and $\mathbb{C}$ carries one imaginary axis, the collinear degeneracy standing as an algebra. One axis cannot host a triad, and the register in which order-sensitivity forces three orthogonal axes over a scalar ground was never entered.

The verification-methodology line. Bayesian confirmation theory, Popperian falsificationism, and Mayo's severe testing each supply a discipline for relating evidence to claims. Each takes the verification apparatus itself as an external presupposition: priors and likelihoods assumed well-formed, the falsifying observation assumed registrable, the severity assessment assumed computable. None derives its own apparatus, none carries a closed-form truth functional, and none exhibits the algebra its compositions live in.

The common structural error is now visible as one sentence. Every fragment inhabits a register that cannot close on itself: the classification has no hypotheses of its own, the determinant has no derivation, the catalog has no selection, the order-sensitivity has no triad, and the methodologies have no apparatus. The assembly point at which all five close simultaneously is the quaternion algebra entered through the architecture's own composition law, and Section 4 performs the assembly.

3. METHODOLOGY • THE TRIAXIAL PROTOCOL AT THE ALGEBRAIC REGISTER

Every load-bearing claim in this paper seals on three warrants. On V_F , the formal-structural axis, each claim is either a named external theorem with its hypotheses exhibited or a computation short enough to be stated whole and reproduced from the text. On V_E , the empirical-thermodynamic axis, each computational claim executes numerically in the embedded instrument of Section 5, with residues stated and a recorded battery any reader can re-run. On V_{ER} , the epistemic-registrational axis, every claim is substrate-portable: reproducible from the statements as printed by any sufficiently capable substrate, with cross-substrate convergence operating on discrete verdicts and parameter-free integer counts, never on scalar entries, per the architecture's portability principle PSP-005 and the standing operator-independent system-role verification record (Islam 2026c).

The warrant phrasing law binds every sentence. Theorem grade attaches only to named theorems and stated computations. Conditional theorem grade attaches where a theorem fires on premises, and the premises travel with the claim. Premise grade attaches to the composition clauses and to substrate monism, each typed to its legislative source. Corroboration grade attaches to witnesses that are removable without motion of any verdict, and Appendix B quarantines all of them. Engineering grade attaches to the recorded battery. The anchor-inflation failure mode, asserting theorem grade where the warrant is conditional, is forbidden, and the licensed phrasings are used exactly.

The premise ledger is short and explicit. CL-1, associativity: iterated audits are bracketing-invariant, demanded by the audit-symmetry legislation, since a verdict economy in which regrouping the same audits changes the outcome is ill-defined. CL-2, integrality: the composite of nonzero warrants is nonzero, demanded by the Mass Mandate with first-failure-terminates, since massive audits cannot compound to a massless verdict. CL-3, linearity with ground identity: composition is $\mathbb{R}$ -bilinear on a finite-dimensional unital carrier and pure ground contact alters no audit, demanded by the evidence pipeline, whose standardization, projection, and Gram evaluation already operate at the linear register. Floor 2, order-sensitivity, premise typed to the operational measurement asymmetry P3: the asymmetry P3 distinguishes the transition from source to target from the transition from target to source for every ordered pair of distinct stations, the operational stations realized as carrier slots once the completion fixes the carrier, and the legislation carries that asymmetry into the composition product, so audit composition is non-commutative. The operational face gates the roster; the algebraic face feeds the Completion. The two faces are one legislative act, not two premises. Floor 1, axis plurality, is deliberately absent from this ledger: Section 4.6 derives it. CL-1 and CL-3 additionally stand

open to direct empirical exercise on recorded session archives, and Section 5.4 states the protocol.

The independence verifiability criterion governs the falsifiable core. The instrument of Section 5 runs on any substrate with floating-point arithmetic, and exact symbolic arithmetic supplies an orthogonal modality for every identity claim: two laboratories, two modalities, no shared failure mode. The integer claims, twenty-four, twelve and twelve, and the class equation, are exhaustively enumerable and parameter-free, the strongest possible decentralization, since a count is checkable by anyone and negotiable by no one.

One ground rule completes the method. The paper proves inline whatever is short, and the single classification too long for the body, the Frobenius theorem, is proved whole in Appendix A. The paper is thereby self-contained for the completion chain, from the composition clauses to the closed-form verdict, modulo three named external consumptions carried at theorem grade: the classification of finite field extensions of $\mathbb{R}$ at Appendix A Step 1, the quantum speed limits at Section 4.1, and Weyl's first fundamental theorem at the catalog. Two further externals, norm multiplicativity and the determinant bound, are retired to inline one-line proofs in Sections 4.4 and 4.9 and consequently drop from the warrant set. Outside those three named consumptions, the citations serve attribution and corroboration, not warrant.

Notation, fixed once. $\mathbb{U}$ is the universal domain. $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$ are the reals, the complexes, the quaternions, with $\text{Im } \mathbb{H}$ the pure quaternions. Evidence rows after standardization and covariate projection are $\tilde{m}_a \in \mathbb{R}^N$ for $a \in \{F, E, ER\}$, with $d_a = \|\tilde{m}_a\|^2/(N - 1)$, unit rows $\hat{q}_a$, correlation Gram R , full Gram G , $\lambda = \text{Re}(\hat{q}_F \hat{q}_E \hat{q}_{ER})$, unit roundoff u_m , collapse floor $\varepsilon = 10^2 \cdot u_m \cdot N$ in double precision and $\varepsilon = 0$ in exact arithmetic. Verdicts are three-state native: $[\checkmark]$ sealed, $[X]$ broken with named mechanism, $[?]$ under-determined.

4. THE CORE DERIVATION

4.1 The Root Axiom and Its Kinetic Floor

The Root Axiom states: $\forall x \in \mathbb{U}, \Delta E_k(x) > 0$. Existence mandates substrate-level kinetic actuation. The axiom enters at axiom grade. It is the ground of the architecture, not a theorem of this paper, and the warrant phrasing law forbids any sentence that pretends otherwise.

What mathematics supplies for the axiom is a floor at the transition register, and it is exact.

Theorem (kinetic floor). Any transition between distinguishable states of a quantum system requires evolution time $\tau \geq \max(\pi\hbar/(2\Delta E), \pi\hbar/(2\langle E \rangle))$, where ΔE is the energy uncertainty and $\langle E \rangle$ the mean energy above the ground state. Consequently every actuation carries a strictly positive energy-time signature. The bound binds unitary, hence logically reversible, processes identically with irreversible ones, since the speed limit is indifferent to logical reversibility. Warrant: theorem grade conditional on the postulates of quantum dynamics, per Mandelstam and Tamm (1945) and Margolus and Levitin (1998). For the irreversible-erasure subclass, Landauer (1961) supplies the sufficient dissipative signature $k_B T \ln 2$ per bit, experimentally confirmed by Bérut and colleagues (2012).

Proposition (triple population, conditional on monism). Any actuation against a target populates three registers: an energetic register through the transition's nonzero energy-time signature, a structural register through the specific transformation selected among alternatives, and a relational register through the coupling by which the actuation reached its target from outside. There is no exterior from which a zero-effect, structureless, boundaryless intervention launches, because such a process would not be an actuation. Warrant: conditional theorem on the single premise of substrate monism, per the Omega Boundary derivation of the standing architecture; the only excluded case is a process leaving no persistent trace, which is by physical fact no actuation.

The static clause, that what exists is substrate configuration and hence subject to the floor whenever it acts, is carried at premise grade on substrate monism. The section closes with its own fence: nothing above is an unconditional mathematical proof of the Root Axiom, and nothing below requires one. The axiom grounds; the floor binds transitions; the derivation now proceeds on the composition law.

4.2 The Composition Law and the Premise Ledger

Composition is native law of the architecture, written into the operational legislation before any algebra is named. Audits compose: a verdict feeds a further audit, evidence superposes, warrants chain. The legislation constrains the composition in three clauses, restated here from Section 3 in their operative form. CL-1: composition is associative, $(A \circ B) \circ C = A \circ (B \circ C)$. CL-2: composition annihilates nothing, the composite of nonzero warrants is nonzero. CL-3: composition is $\mathbb{R}$ -bilinear on a finite-dimensional unital carrier, with the ground acting as two-sided identity, $1 \cdot x = x \cdot 1 = x$. Floor 2: composition is order-sensitive in the two-faced sense legislated in Section 3, its operational face the asymmetry P3 on ordered slot pairs and its algebraic face the non-commutativity of the product, one premise typed to P3.

Each clause is a premise, typed to its legislative source, and each is independently attackable: a recorded session archive in which regrouping identical audits changed a verdict would break CL-1 as instantiated, a composite of massive audits returning the zero record would break CL-2, a failure of additivity at the linear register would break CL-3, and a demonstration that transition direction never carries operational content would break Floor 2. Section 5.4 states the standing empirical protocol. Floor 1, the plurality of axes, is conspicuously not on this list. It will be repaid, not assumed.

4.3 The Wall in Bare Coordinates

Theorem (no triaxial closure). There is no associative unital $\mathbb{R}$ -algebra without zero divisors on a three-dimensional carrier $\text{span}_{\mathbb{R}}\{1, i, j\}$ with $i^2 = j^2 = -1$.

Proof. Closure forces $ij = a + bi + cj$ for some real a, b, c . Associativity gives $i(ij) = (i^2)j = -j$. Expanding the left side: $i(a + bi + cj) = ai + bi^2 + c(ij) = ai - b + c(a + bi + cj) = (ca - b) \cdot 1 + (a + cb) \cdot i + c^2 \cdot j$. Matching the j -coordinate of $-j$ gives $c^2 = -1$, which has no real solution. ■

This is the exact equation that stopped Hamilton's triplets for thirteen years, and it is the entire wall at this register: two lines of coordinate algebra. The three-dimensional triad is not weakly obstructed; it is closed off absolutely. What survives in every dimension is quadratic evaluation; what is quantized is quadratic composition, which by Hurwitz (1898) exists exactly in dimensions 1, 2, 4, 8, a theorem whose proof is itself algebraic. The topological forms of the wall, the combed sphere, the H-space exclusion, the parallelizability classification, corroborate from their own register and are quarantined whole in Appendix B at zero load.

4.4 Scalar Exit and Fertile Orthogonality

Two lemmas govern what a closed multiplicative structure must contain. They are stated first at the composition-algebra register, where (A, n) carries a multiplicative norm, $n(xy) = n(x)n(y)$, with polarization $\langle x, y \rangle = \frac{1}{2}(n(x+y) - n(x) - n(y))$, conjugation $\bar{x} = 2\langle x, 1 \rangle - x$, the standard identities $\langle xy, z \rangle = \langle y, \bar{x}z \rangle$, $\langle ux, uy \rangle = n(u)\langle x, y \rangle$, $\langle xu, yu \rangle = \langle x, y \rangle n(u)$, and the rank equation $x^2 - 2\langle x, 1 \rangle x + n(x) = 0$, all classical (Springer and Veldkamp 2000). An element is pure when $\langle x, 1 \rangle = 0$.

Lemma (Scalar Exit). A pure unit u satisfies $u^2 = -1$. Self-composition exits the axis space onto the scalar line; a triad closed under its own products must carry a scalar slot.

Proof. The rank equation with $\langle u, 1 \rangle = 0$ and $n(u) = 1$ reads $u^2 + 1 = 0$. ■

Lemma (Fertile Orthogonality). For orthogonal pure units $u \perp v$, the product uv is a pure unit orthogonal to 1, to u , and to v . Hence the multiplicatively closed subspace containing two orthogonal axes contains the four-dimensional span $\{1, u, v, uv\}$, and three slots can never close. (Closure of that span as a subalgebra at the composition-algebra register follows from alternativity: composition algebras are alternative, so by Artin's theorem the subalgebra generated by u and v is associative and equals span $\{1, u, v, uv\}$; inside $\mathbb{H}$ this is immediate.)

Proof. $n(uv) = n(u)n(v) = 1$. Purity: $\langle uv, 1 \rangle = \langle v, \bar{u} \rangle = \langle v, -u \rangle = 0$. Second orthogonality: $\langle uv, u \rangle = \langle uv, u \cdot 1 \rangle = n(u)\langle v, 1 \rangle = 0$. Third: $\langle uv, v \rangle = \langle uv, 1 \cdot v \rangle = \langle u, 1 \rangle n(v) = 0$. ■

Hypothesis hygiene, stated once. These lemmas run pre-completion at the composition-algebra register, where the multiplicative norm is hypothesis. After Section 4.6 they hold unconditionally inside $\mathbb{H}$, whose norm is multiplicative by the one-line identity $n(uv) = (uv)(uv)^- = uv \bar{v}\bar{u} = u n(v) \bar{u} = n(u)n(v)$ from the conjugation anti-automorphism $(uv)^- = \bar{v}\bar{u}$, which is the inline retirement of Hurwitz from every bound below. There the second lemma is one line: for pure u, v the product law $uv = -\langle u, v \rangle + u \times v$ makes the scalar part vanish under orthogonality and the vector part the cross product, orthogonal to both factors. Nothing downstream consumes the lemmas outside $\mathbb{H}$.

Together the lemmas fix the anatomy in advance of the classification: one axis is sterile, its self-composition collapsing onto the scalar line; two orthogonal axes beget a third and force a four-slot carrier. Fertility and closure jointly demand the shape $1 + 3$ before any theorem names it.

4.5 The Order-Sensitivity Completion

Definition. In a finite-dimensional associative unital $\mathbb{R}$ -division algebra A , the ground is $\mathbb{R} \cdot 1$; the trace-zero space is $V(A) = \{x \in A : x^2 \in \mathbb{R}, x \notin \mathbb{R}\} \cup \{0\}$,

which Appendix A shows to be a linear complement of the ground; the axis count of A is $\dim V(A) = \dim A - 1$.

Theorem (Order-Sensitivity Completion). Let A satisfy CL-1, CL-2, CL-3. The following are equivalent. (i) Composition in A is order-sensitive: A is non-commutative. (ii) A carries at least two linearly independent axes. (iii) $A \cong \mathbb{H}$. (iv) The axis count of A is exactly three.

Proof. By CL-3, A is a finite-dimensional unital $\mathbb{R}$ -algebra; by CL-1 it is associative; by CL-2 it has no zero divisors, and in finite dimension this makes every nonzero element invertible, so A is a division algebra. By the Frobenius classification, proved self-contained in Appendix A, A is isomorphic to $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$, with axis counts 0, 1, 3 respectively, $\mathbb{R}$ and $\mathbb{C}$ commutative and $\mathbb{H}$ non-commutative. (i) $\implies$ (iii): non-commutativity excludes $\mathbb{R}$ and $\mathbb{C}$, leaving $\mathbb{H}$. (ii) $\implies$ (iii): axis count at least two excludes counts 0 and 1, leaving $\mathbb{H}$. (iii) $\implies$ (iv): $\dim \text{Im } \mathbb{H} = 3$. (iv) $\implies$ (ii): immediate. (iii) $\implies$ (i): $ij = k \neq -k = ji$. The four statements loop. ■

The theorem is the spine of the paper, and its content is the unification of two demands the architecture carries separately at its other registers. The directed roster presupposes that the transition $(s \rightarrow t)$ is a different operational object from $(t \rightarrow s)$; operationally this is the asymmetry P3 that has always directed the gates, Floor 2's operational face; algebraically it is clause (i), Floor 2's algebraic face; the two are one premise by the Section 3 legislation, not an identification asserted here. The plurality of axes is clause (ii). The theorem makes them faces of one premise under the classification: the same order-sensitivity that directs the gates forces the triaxial count, and neither demand is separable from the other again. One premise in, two structures out, no parameter free.

4.6 Triaxial Forcing and the Corollary Floor

Theorem (Triaxial Forcing). Under CL-1, CL-2, CL-3 and order-sensitivity, the verification algebra is the quaternions, $A \cong \mathbb{H} = \mathbb{R} \oplus \text{Im } \mathbb{H}$: one scalar ground, exactly three orthogonal axes. The octonions fall to CL-1, their product non-associative; every Cayley-Dickson stage past the octonions falls to CL-2, the sedenions onward carrying zero divisors (Zorn 1933 for the alternative classification; the sedenion zero divisors by direct construction).

Proof. Section 4.5, (i) $\implies$ (iii), with the exclusions as cited. ■

Corollary (Floor 1, derived). At least two linearly independent verification axes exist; indeed exactly three. The plurality floor, carried in the master codex's completion as an explicit premise of its Section 1A, is at this register a corollary of order-sensitivity under the classification. The premise ledger of the architecture shrinks by one.

Corollary (no fourth axis). A fourth orthogonal axis would demand a five-dimensional associative division carrier, and none exists: the Frobenius inventory contains no dimension five, just as it contains no dimension three. The next dimension carrying composition of any kind is eight, and the octonions fall to CL-1. The triaxial count is capped above and below by the same classification.

4.7 The Involution, the Center, and the Registration Line

The completion carries the triad as structure, not as stipulation, and three theorems make the carriage exact.

Theorem (conjugation eigenspace). Quaternion conjugation $q \mapsto \bar{q}$ is an involutive anti-automorphism of $\mathbb{H}$ whose $+1$ eigenspace is the ground $\mathbb{R}$ and whose -1 eigenspace is the axis space $\text{Im } \mathbb{H}$. The triad is the negative eigenspace.

Proof. Conjugation fixes $\mathbb{R}$, negates $\text{Im } \mathbb{H}$, reverses products by direct computation on the basis, and squares to the identity; the eigenspace split is the canonical decomposition $\mathbb{H} = \mathbb{R} \oplus \text{Im } \mathbb{H}$. ■

Theorem (uniqueness at the ground). Conjugation is the unique $\mathbb{R}$ -linear involutive anti-automorphism of $\mathbb{H}$ whose fixed subspace is exactly the ground $\mathbb{R}$.

Proof. Let τ be such a map. First $\tau(1) = 1$, since $\tau(1) = \tau(1 \cdot 1) = \tau(1)^2$ and $\tau(1)$ is invertible. Since $\tau^2 = \text{id}$ and τ is $\mathbb{R}$ -linear, $\mathbb{H}$ splits as $E_+ \oplus E_-$, the ± 1 eigenspaces of τ , with $E_+ = \text{Fix}(\tau) = \mathbb{R}$ by hypothesis, so $\dim E_- = 3$. For $x \in E_-$: $\tau(x^2) = \tau(x)\tau(x) = (-x)(-x) = x^2$, so $x^2 \in \text{Fix}(\tau) = \mathbb{R}$. If $x \neq 0$ and $x^2 = s \geq 0$, then $(x - \sqrt{s})(x + \sqrt{s}) = 0$, and the division property forces $x = \pm\sqrt{s} \in \mathbb{R} \cap E_- = \{0\}$, a contradiction; hence $x^2 < 0$, so x is trace-zero and $E_- \subseteq \text{Im } \mathbb{H}$, with equality by dimension. Then τ fixes $\mathbb{R}$ pointwise and negates $\text{Im } \mathbb{H}$, which is conjugation. ■

Remark (why the qualifier is load-bearing). Involutions of orthogonal type, $x \mapsto u \bar{x} u^{-1}$ for a pure unit u , are also involutive anti-automorphisms fixing the ground pointwise, but each fixes a three-dimensional subspace: for $u = i$ the fixed space is $\text{span}\{1, j, k\}$. The characterizing property of conjugation among all involutions of the first kind is that its fixed subspace is exactly the ground, the symplectic type in the standard classification of involutions (Knus, Merkurjev, Rost, and Tignol 1998). The triad is therefore not the negative eigenspace of some involution; it is the negative eigenspace of the only involution whose symmetric elements are precisely the registration line.

Theorem (center). $Z(\mathbb{H}) = \mathbb{R}$.

Proof. Let $z = a + bi + cj + dk$ commute with i : $zi = -b + ai + dj - ck$ and $iz = -b + ai - dj + ck$, so $c = d = 0$. Then $z = a + bi$; commuting with j gives $zj = aj + bk$ and $jz = aj - bk$, so $b = 0$ and $z = a \in \mathbb{R}$. The reverse containment is immediate. ■

The center is the one subspace commuting with every audit, which is what a registration line must be: a record that composition from either side cannot deform. The theorem supplies the uniqueness of that commuting line; the naming of the line as the Root Axiom's registration coordinate is the architecture's convention, premise-typed as a definitional identification and never claimed as a theorem.

4.8 The Twelve in Pure Algebra

The completed carrier has four canonical slots: the ground and the three axis directions of any orthonormal axis frame, the frame choice a gauge that Section 4.9 discharges by conjugation invariance. Floor 2 gates the ordered pairs of distinct slots and gates all of them. Three witnesses now deliver the cardinality, each inside algebra, none touching geometry.

Witness W1, slot combinatorics. Ordered pairs of distinct slots among four: $4 \cdot 3 = 12$. One line, conditional on the forced carrier of Section 4.6 and on Floor 2, both already in the ledger. W1 consumes the universal face of Floor 2, that every ordered pair is distinct and gated; the Completion of Section 4.5 consumes only the existential face, that some pair fails to commute; both faces are the one P3 legislation, and the stronger universal reading is the operative premise.

Theorem (W2, the torsor). A_4 is the unique subgroup of S_4 of order twelve, and it acts simply transitively on the twelve ordered pairs of distinct letters.

Proof. Uniqueness: a subgroup of index two is normal with quotient $\mathbb{Z}_2$, hence contains every square; every 3-cycle is a square, $(abc) = (acb)^2$, and the 3-cycles generate A_4 , so the subgroup contains A_4 and order forces equality. Simple transitivity: the stabilizer in A_4 of the ordered pair (a, b) consists of even permutations fixing a and b pointwise, hence even elements of the symmetric group on the remaining two letters; the only candidates are the identity and the transposition of those letters, the transposition is odd, and the stabilizer is trivial. Orbit-stabilizer gives an orbit of size $|A_4| = 12$, the whole set. ■

The roster is an A_4 -torsor: between any two gates exactly one symmetry, no gate canonical, the cascade's entry point a choice of base and never a privileged element. The proof is pure permutation algebra; no rotation, no polyhedron, no metric enters it.

Theorem (the double cover). The unit group of the Hurwitz order in $\mathbb{H}$ consists of exactly twenty-four elements: the eight axis units $\pm 1, \pm i, \pm j, \pm k$ and the sixteen diagonal units $(\pm 1 \pm i \pm j \pm k)/2$. It is closed under multiplication with every element of unit norm, it contains $\{\pm 1\}$ centrally, and the quotient by $\{\pm 1\}$ is a group of order twelve whose conjugacy classes of antipodal pairs have sizes 1, 3, 4, 4, the class equation of A_4 ; the quotient is isomorphic to A_4 , the binary tetrahedral identification (Conway and Smith 2003). Section 5 exhibits

the count, the closure, the norms, and the class equation by exhaustive enumeration. The cardinality of the cascade lives inside the integers of $\mathbb{H}$, doubled by the spin kernel.

Theorem (W3, shell arithmetic). The Hurwitz integers of norm 2 number exactly twenty-four; they split twelve and twelve by occupancy of the real slot; the pure-imaginary slice numbers exactly twelve.

Proof. A half-integer Hurwitz element has all four coordinates odd halves $o_m/2$ with o_m odd; its norm is $(o_1^2 + o_2^2 + o_3^2 + o_4^2)/4$, each odd square is $\equiv 1 \pmod{8}$, the numerator is $\equiv 4 \pmod{8}$, and the norm is odd, so no half-integer element has norm 2. The norm-2 elements are therefore the integer solutions of $a^2 + b^2 + c^2 + d^2 = 2$: exactly two coordinates equal to ± 1 and two equal to zero, counted by $C(4,2) \cdot 2^2 = 24$. The real slot is occupied in $3 \cdot 4 = 12$ of them and zero in $C(3,2) \cdot 2^2 = 12$; the latter twelve constitute the pure-imaginary slice. ■

Corroboration from arithmetic: Jacobi's four-square count gives $r_4(2) = 8\sigma(2) = 24$ (Jacobi 1829). The geometric reading of the slice as the face-centered-cubic kissing configuration of three-dimensional space stands in Appendix B at zero load.

Direction as sign. The axis-axis sextet of gates runs between imaginary directions, where order is orientation: $ij = k$ while $ji = -k$, the anticommutation $ij = -ji$ carrying directedness as sign inside the algebra itself. The seal-incident sextet runs against the center, which commutes; the directedness of those six is carried by Floor 2's operational face through the role typing of the Operational Content Theorem, the same premise that directs the axis-axis sextet by sign, not a separate import.

The fence, restated as standing law. The algebra carries the cardinality, the symmetry group, the double cover, the shell arithmetic, and the orientation law of the roster. The content of each gate, which pathology each ordered pair prevents, is forced by the Operational Content Theorem on source-role and target-role pairings and is never derived from the algebra. This paper derives twelve-ness three ways; it imports the twelve gates by reference.

4.9 The Closed-Form Truth Function

The pipeline objects are fixed in Section 3: standardized evidence rows, covariate projection under the regularity conditions, residue rows $\tilde{m}_a$, scale factors d_a , unit rows $\hat{q}_a$, correlation Gram R , full Gram G .

Theorem (Triple-Product Verdict Identity). Let $\hat{q}_F, \hat{q}_E, \hat{q}_{ER}$ be the unit residue rows in $\mathbb{R}^N$, $N \geq 3$, with correlation Gram R . Under any linear isometry of their span into $\text{Im } \mathbb{H}$, with images written by the same names, $\det(R) = (\text{Re}(\hat{q}_F \hat{q}_E \hat{q}_{ER}))^2 = \lambda^2$.

Proof. For pure quaternions the product law is $uv = -\langle u, v \rangle + u \times v$, hence $\text{Re}(uvw) = -\langle u \times v, w \rangle = -\det(u, v, w)$. With M the 3×3 coordinate matrix of the rows inside their own span, $\det(R) = \det(MM^T) = \det(M)^2 = \lambda^2$. The identification is unique up to an orthogonal map of the span; an orientation-reversing component flips the sign of λ , the square is invariant, and the sign routes to the orientation annotation register out-of-band. If the span has dimension below three, the rows are dependent, $\det(R) = 0$, and the composition is pure with $\lambda = 0$. ■

Theorem (pipeline factorization). $\det(G) = d_F d_E d_{ER} \cdot \det(R)$, at identical sign and zero set.

Proof. $G = D^{\{1/2\}} R D^{\{1/2\}}$ with $D = \text{diag}(d_F, d_E, d_{ER})$; take determinants. ■

The identity's content is the connection named in Section 1: the statistician's volume is the squared scalar residue of the quaternionic composition. Write $w = \hat{q}_F \hat{q}_E \hat{q}_{ER} = \cos \theta + \hat{n} \sin \theta$. The verdict is the squared cosine of the composed triad against the scalar line, $\det(R) = \cos^2 \theta$, every verdict state a position of the composition against the ground.

The bounds are derived inline, with no external citation. $|\lambda| \leq 1$ by the norm-multiplicativity identity of Section 4.4 on unit factors, so $\det(R) = \lambda^2 \in [0, 1]$. For the pipeline face, R is a correlation matrix of trace 3 with nonnegative eigenvalues $\lambda_1, \lambda_2, \lambda_3$ summing to 3, and the identity $x^3 + y^3 + z^3 - 3xyz = (x + y + z) \cdot \frac{1}{2}((x - y)^2 + (y - z)^2 + (z - x)^2) \geq 0$ at $x = \lambda_1^{1/3}, y = \lambda_2^{1/3}, z = \lambda_3^{1/3}$ gives $\lambda_1 \lambda_2 \lambda_3 \leq ((\lambda_1 + \lambda_2 + \lambda_3)/3)^3 = 1$, so $\det(R) \leq 1$, and $\det(G) = (\prod d_a) \det(R) \leq \prod d_a \leq 1$ by the factorization with each $d_a \in [0, 1]$ under the non-expansive projection; Hadamard is thereby retired from the warrant set. The norm is never the verdict: a fully collinear unit triad returns $\det(R) = 0$ while the composed norm sits pinned at one, the norm blind to collapse, its one verdict role the ceiling.

Maximal lock is the Hamilton relation: $\det(R) = 1$ exactly when the triad is orthonormal, and then the composition lands on the scalar line at $\lambda = \mp 1$, in the right-handed frame the carved relation itself, $ijk = -1$. Breakage is fourth-axis genesis: collinear and coplanar triads compose pure, $\lambda = 0$, $w^2 = -1$, the collapsed pair consuming itself into the scalar and leaving the third axis bare, $\hat{q} \hat{q}_{ER} = -\hat{q}_{ER}$. Redundancy is not partial credit but broken geometry.

Theorem (catalog completeness). Every rotation-invariant real polynomial functional of the triad is a polynomial in the Gram entries and λ ; equivalently, the entire catalog of admissible invariant truth functionals lives in the real-part subring of quaternion words.

Proof. Weyl's first fundamental theorem for the rotation group generates the invariants of vector tuples in $\mathbb{R}^3$ by pairwise inner products and 3×3

determinants, and for pure quaternions both generators are real parts of words: $\langle u, v \rangle = -\text{Re}(uv)$ and $\det(u, v, w) = -\text{Re}(uvw)$. ■

The verdict functional is therefore not one permitted invariant among unknown others: the catalog is closed, the seal generates it, and any frame-dependent alternative measures the analyst's coordinates rather than the proposition.

Theorem (frame invariance). Rotating the entire axis frame is conjugation by a unit quaternion r : each pure axis maps to $r \hat{q} \bar{r}$, which remains pure, the composition transforms as $w \mapsto r w \bar{r}$ by associativity, and $\text{Re}(r w \bar{r}) = \text{Re}(w \bar{r} r) = \text{Re}(w)$, since the real part of a quaternion product is invariant under cyclic exchange. ■

Precedence is legislated, not derived, and is stated exactly. Admissibility [?] first: the dimensional floor $N \geq k + 4$, nonzero row variance, covariate rank k , $\kappa(\tilde{C}\tilde{C}^T) < 10^6$, with the constant carried as covariate zero in centering form. Collapse [X] second: $\det(R) \leq \varepsilon$. Conditioning ceiling [?] third: $\kappa(G) \geq 10^6$ certifies only the positive branch. The remainder seals [⌞]. Collapse outranks conditioning, and Section 5 exhibits every branch, including an exactly collapsed configuration whose condition number cannot intercept the [X].

4.10 The Return Law

The Return is scalar contact: $\lambda \neq 0$, the composed triad projecting nonzero onto the ground line. The full Return is the Hamilton landing: $w = \pm 1$, $\det(R) = 1$, the composition the scalar itself. The dichotomy at the boundary is exact, scalar contact against pure axis, $\lambda \neq 0$ against $\lambda = 0$ with $w^2 = -1$. The ground is the line; the seal is the touch. The Master Unknotting holds as an equation, $\text{GOL} \iff \lambda \neq 0$, where GOL, the Geometric Orthogonal Lock, is the architecture's name for the cascade verdict event, here the event of scalar contact with the registration line; it is the closed form of APEX-PSP-MU-01, and at this register it is one line of Section 4.9: the lock event and the nonvanishing of the scalar residue are the same predicate.

5. FALSIFIABLE CONTENT · THE INSTRUMENT AND THE RECORDED BATTERY

5.1 The Reference Instrument

Any substrate with floating-point arithmetic loads and runs the following. It is the executable form of the verdict pipeline and the closed-form identity jointly, with the precedence of Section 4.9 implemented exactly: admissibility before collapse, collapse before conditioning.

```
```python import numpy as np
```

```
def qmul(a, b): w1,x1,y1,z1 = a; w2,x2,y2,z2 = b return np.array([w1w2-x1x2-y1y2-z1z2, w1x2+x1w2+y1z2-z1y2, w1y2-x1z2+y1w2+z1x2, w1z2+x1y2-y1x2+z1w2])
```

```
def trisduct(M, C=None, exact=False): M = np.asarray(M, float); N = M.shape[1] Cm = None if C is None else np.atleast_2d(np.asarray(C, float)) k = 0 if Cm is None else Cm.shape[0] u_m = np.finfo(float).eps eps = 0.0 if exact else 100.0u_mN if N - k < 4: return '[?]', None, None, None, 'N-k<4 dimensional shortfall' Mn = M - M.mean(axis=1, keepdims=True) sd = Mn.std(axis=1, ddof=1, keepdims=True) if np.any(sd == 0): return '[?]', None, None, None, 'zero-variance row' Mn = Mn / sd if k: Cm = Cm - Cm.mean(axis=1, keepdims=True) # Route A: constant as covariate zero, centering form if np.linalg.matrix_rank(Cm) < k: return '[?]', None, None, None, 'rank(C)<k' CC = Cm @ Cm.T if np.linalg.cond(CC) >= 1e6: return '[?]', None, None, None, 'kappa(CC^T)>=1e6' Mf = Mn - (Mn @ Cm.T) @ np.linalg.solve(CC, Cm) else: Mf = Mn d = (MfMf).sum(axis=1) / (N-1) G = Mf @ Mf.T / (N-1) detG = float(np.linalg.det(G)) if np.any(d <= eps): lam, detR = 0.0, 0.0 else: Q = Mf / np.sqrt((MfMf).sum(axis=1, keepdims=True)) R = Q @ Q.T detR = float(np.linalg.det(R)) B = np.linalg.svd(Q, full_matrices=False)[2][:3] co = Q @ B.T q = [np.concatenate([0.0, c]) for c in co] lam = float(qmul(qmul(q[0], q[1]), q[2]))[0]) if detR <= eps: return '[X]', lam, detR, detG, 'collapse: det(R)<=eps' if np.linalg.cond(G) >= 1e6: return '[?]', lam, detR, detG, 'kappa(G)>=1e6' return '[✓]', lam, detR, detG, 'sealed' ```
```

### 5.2 The Recorded Battery

Sixteen checks, executed at forge time on 11 June 2026 in IEEE-754 double precision, numpy 2.4.4, seed 20260611, reproducible from the statements as printed. Identity checks report residues; enumerative checks report exact integers; verdict checks report the branch taken.

---

M-CHK.1, identity sweep: five hundred random unit triads in  $\mathbb{R}^3$ , maximal  $|\lambda^2 - \det R| = 4.2 \times 10^{-16}$ . M-CHK.2, identity under span isometry: two hundred trials with row length  $N$  drawn from  $[4, 40)$ , maximal deviation  $2.8 \times 10^{-15}$ ,  $\det R$  confined to  $[0, 1]$  with observed range  $[3.1 \times 10^{-2}, 0.998]$ . M-CHK.3, factorization sweep: two hundred full-pipeline trials,  $N \in [8, 40]$ ,  $k \in [0, 3]$ , maximal  $|d_F d_E d_{ER} \cdot \det R - \det G| = 8.9 \times 10^{-16}$ , maximal  $\det G = 0.9913$  under the unit ceiling of Section 4.9, sign and zero set of  $\det G$  and  $\det R$  in agreement on every trial. M-CHK.4, handedness: the right-handed orthonormal triad returns  $\lambda = -1.000000000000$  and the left-handed  $\lambda = +1.000000000000$ , with  $\lambda^2 = 1 = \det R$  both. M-CHK.5, sealed control at  $N = 20$  with two mass covariates:  $[\mathbf{v}]$ ,  $\lambda = -0.803446614094$ ,  $\det R = 0.645526461698$ ,  $\det G = 0.328130421577$ ,  $|\lambda^2 - \det R| = 6.7 \times 10^{-16}$ , and the factorization residue evaluating to zero at double precision on this configuration. M-CHK.6, collinear triad:  $[X]$  collapse, with  $\kappa(G) = 2.1 \times 10^{16}$  on the same configuration intercepted by nothing, the precedence construction demanded by the load checks. M-CHK.7, coplanar triad:  $[X]$ ,  $\lambda = -2.3 \times 10^{-17}$ , composition pure-imaginary with  $w^2 = (-1, 0, 0, 0)$  to machine precision. M-CHK.8, inadmissibility: dimensional shortfall  $N - k < 4$  routes  $[?]$ , and an ill-conditioned covariate block routes  $[?]$  at  $\kappa(\tilde{C}\tilde{C}^T)$ ; the zero-variance and rank-deficient branches route  $[?]$  with their mechanisms named. M-CHK.9, Hamilton landing on a centered orthonormal triad:  $[\mathbf{v}]$ ,  $\det R = 1.000000000000$ ,  $|\lambda| = 1.000000000000$ . M-CHK.10, invariance: evidence-coordinate relabel by column permutation moves  $\det R$  by  $2.2 \times 10^{-16}$ ,  $\lambda^2$  by  $7.8 \times 10^{-16}$ ,  $\det G$  by zero; conjugation of the unit triad by a random unit quaternion moves  $\lambda$  by  $4.4 \times 10^{-16}$ . M-CHK.11, choice-independence: proper and improper span identifications on one fixed configuration return  $\lambda = -0.803446614094$  and  $\lambda = +0.803446614094$ , the square fixed to the last digit. M-CHK.12, Hurwitz unit group: exhaustive enumeration returns twenty-four elements, closed under multiplication, every element of unit norm. M-CHK.13, class equation: the antipodal pairs of the unit group fall into conjugacy classes of sizes 1, 3, 4, 4. M-CHK.14, the torsor: the twelve even permutations of four letters act on the twelve ordered pairs in a single orbit with every stabilizer trivial, by exhaustive check. M-CHK.15, shell arithmetic: the norm-2 integer quaternions number twenty-four, the pure-imaginary slice twelve, the real-occupied complement twelve, and the half-integer norm scan returns attainable values 1 and 3 near the target with 2 absent. M-CHK.16, the norm law: five hundred random unit pairs return maximal  $|\|uv\| - 1| = 3.3 \times 10^{-16}$ , and the self-consumption relation  $u u v = -v$  holds to  $2.2 \times 10^{-16}$ .

### 5.3 Falsifier Clauses

Each clause states the prediction, the confirming modality, the expected signal, and the null result that breaks the claim, with no wiggle room.

**F1, the verdict identity.** Prediction:  $\lambda^2 = \det R$  on every admissible triad. Modalities: double-precision execution of the instrument, and exact symbolic arithmetic as the orthogonal channel. Expected: residues bounded by a small multiple of  $u_m \cdot N$  in floating point and exactly zero in exact arithmetic. Null: a single admissible configuration with a residue persistently exceeding  $10^{-12}$  at double precision, or any exact-arithmetic counterexample, falsifies the identity claim.

**F2, the factorization.** Prediction:  $\det G = d_F d_E d_{ER} \cdot \det R$  at identical sign and zero set on every pipeline configuration. Modalities and thresholds as F1. Null: any configuration separating the sign or zero set of the two faces falsifies the factorization.

**F3, the integer core.** Prediction: the Hurwitz unit group has exactly twenty-four elements and is multiplicatively closed; the antipodal-pair class equation is 1, 3, 4, 4;  $A_4$  has order twelve and acts on the twelve ordered pairs with a single orbit and trivial stabilizers; the norm-2 shell has exactly twenty-four elements splitting twelve and twelve with no half-integer member. Modality: exhaustive enumeration, checkable by hand. Null: any deviation in any count falsifies the corresponding theorem as stated; these are parameter-free integers and admit no tolerance.

**F4, the invariances.** Prediction: column relabel and frame conjugation leave  $\det R$ ,  $\lambda^2$ , and  $\det G$  fixed to machine precision; improper span identification flips the sign of  $\lambda$  and fixes the square. Null: any invariance breach beyond the F1 threshold falsifies the corresponding invariance theorem as implemented.

**F5, the precedence.** Prediction: each of  $[\kappa]$ ,  $[X]$ ,  $[?]$  is constructively reachable, and an exactly collapsed configuration returns  $[X]$  regardless of its condition number. Null: a collapsed configuration returning any verdict other than  $[X]$ , or an unreachable branch, falsifies the precedence implementation.

**F6, the bounds.** Prediction:  $\det R$  and  $\det G$  lie in  $[0, 1]$  on every admissible configuration. Null: any escape beyond numerical tolerance falsifies the bound derivation as implemented.

**F7, the composition law.** Prediction: on recorded session archives, regrouping identical audits leaves verdicts invariant and warrant superposition is additive at the linear register; composites of nonzero-warrant records are nonzero. Null: a reproducible archive violating bracketing invariance, additivity, or integrality breaks the corresponding clause as instantiated, and with it the conditional theorems of Section 4 at their premise.

Failure of any check of Section 5.2 on independent re-execution falsifies the corresponding closed-form claim, the standing instrumentation-falsification discipline of the architecture applied to this paper whole.

## 5.4 The Composition Law Under Empirical Exercise

CL-1 and CL-3 are not merely premises with legislative pedigrees; they are exposed. The standing protocol reads them off recorded cascade archives: compose recorded audits under both bracketings and under content superposition, and read verdict invariance as the empirical signature of CL-1 and additivity of  $\lambda$  at the linear register as the signature of CL-3. CL-2 is exercised by composing nonzero-warrant records and confirming that no composite of nonzero- $\lambda$  audits returns the zero record; the master codex's instrumentation ledger records one thousand such compositions returning minimum composite magnitude 1.000000000000 at unit factors, and M-CHK.16 above exercises the same norm law on five hundred fresh pairs. Floor 2 is exposed operationally: a demonstration that transition direction never carries operational content would break it. The premises of this paper are therefore not protected assumptions; they are standing experimental targets, and the conditional theorems built on them inherit exactly that exposure.

## 6. DISCUSSION · WHAT THE THIRD SEAL CLOSES

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The independence law is now satisfied at its third face. Seal L runs with no geometry and no algebra; Seal G runs with no multiplication; this paper runs Seal M with no topology and no linguistics, and the verdicts do not move. The three fences are mirror images: each seal quarantines the other two instrument sets as corroboration, and the convergence of the three on one verdict economy is exhibited rather than assumed. A reader who distrusts deletion tests and distrusts polyhedra can now enter the architecture through algebra alone and re-derive the completion chain from the composition law and Appendix A.

The premise ledger shrinks, and the shrink is structural. The completion of the master codex consumes four inputs at its own register: three composition clauses and a plurality floor. The Order-Sensitivity Completion shows the floor is a face of the directedness the roster already demanded, so the ledger at this register reads CL-1, CL-2, CL-3, Floor 2, with monism carried separately for the floor proposition of Section 4.1 and the Root Axiom standing as the ground. Every remaining premise is typed, attackable, and partially under standing empirical exercise. A verification architecture whose premises are experimental targets is in a different epistemic position from one whose premises are commitments.

The gauge clause deserves its own statement. The labeling of the three imaginary units to the three verification axes is conventional: the automorphisms of  $\mathbb{H}$  are inner and act as the full rotation group on  $\text{Im } \mathbb{H}$ , the verdict functional is conjugation-invariant by the frame-invariance theorem, and the battery exhibits the invariance at machine precision together with the sign flip under improper identification, the sign routing out-of-band to the orientation annotation register. The sealed content is the count and the invariant functional, never the labels. Cross-substrate comparison therefore operates exactly where it should: on the discrete verdict, the squared invariant, and the integer core.

One algebra carries two registers, and the derivation keeps them separate. Registrational separation is linear-register law, where the Gram discipline operates and where the verdict reads a determinant. Generative fertility is multiplicative-register law, where orthogonal axes beget:  $k = ij$  while  $k \notin \text{span}_{\mathbb{R}}\{1, i, j\}$ , multiplicative generation and linear irreducibility simultaneous in one algebra, the joint demand that forces dimension four. Verification never multiplies, generation never substitutes, and the closed-form identity is precisely the bridge on which the two registers meet without merging.

The failure surfaces are named honestly. The cited classifications are theorems with proofs reproduced or referenced; they do not fail. What can fail is exhaustively listed: the implementation, exposed by Section 5.2 and falsified by

F1 through F6; the premises, exposed by Section 5.4 and falsified by F7; and the imports, the Operational Content Theorem at operational warrant and the quantization mapping at engineering grade, each typed and each outside this paper's claims. An attack on the architecture at the mathematical register must land on one of these surfaces, and the surfaces are printed.

Adjacent structures become tractable from here. The exact geometry of the orientation sign, held out-of-band, is a developed annotation register in the master codex and inherits the present closed form. The standing open question this paper sharpens rather than settles is the uniqueness of gate content: the Operational Content Theorem forces the directed-edge-to-content bijection at operational warrant, and whether an operational-theoretic uniqueness proof of independent strength exists is the natural next target, with the fence of Section 4.8 marking exactly where algebra stops and that work begins.

The limits, restated once and plainly. The Root Axiom is an axiom. The kinetic floor is conditional on quantum dynamics. The triple-population proposition is conditional on monism. The completion theorems are conditional on the composition clauses. The gate content is imported. The collapse floor  $\varepsilon$  is numerical-analysis engineering. Nothing in this paper outruns these statements, and the warrant ledger of Appendix C carries every claim with its tier.

## 7. CONCLUSION

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The question this paper opened is whether the mathematical seal of the Trisduction architecture stands alone, with the linguistic and geometric instrument sets removed entirely. It does. From the Root Axiom at axiom grade and its quantum-speed-limit floor at conditional theorem grade, through the architecture's own composition law, the Order-Sensitivity Completion forces the quaternions, the triaxial count, and the corollary plurality floor; the twelve-gate cardinality is recovered three ways inside the integers of  $\mathbb{H}$ ; and the truth function closes as  $\det(R) = (\text{Re}(\hat{q}_F \hat{q}_E \hat{q}_{ER}))^2$  with derived bounds, a closed invariant catalog, conjugation frame-invariance, explicit precedence, and the Return Law  $\text{GOL} \iff \lambda \neq 0$ .

The primary falsifiable content is printed and recorded: a reference instrument, a sixteen-check battery with residues at machine precision, and an integer core, twenty-four, twelve and twelve, and 1, 3, 4, 4, that admits no tolerance and no negotiation. The falsifier clauses F1 through F7 state the exact null results that would break each claim.

The execution call is unusually cheap to answer. Any substrate with floating-point arithmetic, human or synthetic, institutional or solitary, loads the instrument from this document and re-runs the battery; exact arithmetic supplies the orthogonal modality; the enumerations are checkable by hand. No observatory is required. The decentralization is total because the claims are algebraic.

The single most important open question is the one the fence protects: whether the operational content of the twelve gates admits a uniqueness proof of strength comparable to the cardinality results, at its own operational register, without ever drawing content from the algebra.

Acceptance of this paper's result entails one reframing, stated in a sentence: the verification algebra of the architecture is not chosen but forced, and its truth functional is not designed but read off, the unique invariant residue of composed evidence on the one carrier that order-sensitive composition permits.

## 8. REFERENCES

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- Adams, J. F. 1960. On the non-existence of elements of Hopf invariant one. *Annals of Mathematics* 72: 20-104.
- Bartlett, M. S. 1951. The effect of standardization on a chi-squared approximation in factor analysis. *Biometrika* 38: 337-344.
- Bérut, A., A. Arakelyan, A. Petrosyan, S. Ciliberto, R. Dillenschneider, and E. Lutz. 2012. Experimental verification of Landauer's principle linking information and thermodynamics. *Nature* 483: 187-189.
- Bott, R., and J. Milnor. 1958. On the parallelizability of the spheres. *Bulletin of the American Mathematical Society* 64: 87-89.
- Busemeyer, J. R., and P. D. Bruza. 2012. *Quantum Models of Cognition and Decision*. Cambridge: Cambridge University Press.
- Conway, J. H., and D. A. Smith. 2003. *On Quaternions and Octonions: Their Geometry, Arithmetic, and Symmetry*. Natick: A K Peters.
- Ebbinghaus, H.-D., H. Hermes, F. Hirzebruch, M. Koecher, K. Mainzer, J. Neukirch, A. Prestel, and R. Remmert. 1991. *Numbers*. Graduate Texts in Mathematics 123. New York: Springer.
- Frobenius, F. G. 1878. Über lineare Substitutionen und bilineare Formen. *Journal für die reine und angewandte Mathematik* 84: 1-63.
- Hadamard, J. 1893. Résolution d'une question relative aux déterminants. *Bulletin des Sciences Mathématiques* 17: 240-246.
- Hamilton, W. R. 1844. On quaternions; or on a new system of imaginaries in algebra. *Philosophical Magazine*, third series, 25: 489-495.
- Hurwitz, A. 1898. Über die Composition der quadratischen Formen von beliebig vielen Variabeln. *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse*: 309-316.
- Islam, M. F. 2026a. On the Topology of Theories of Everything. PhilArchive record ISLTOT.
- Islam, M. F. 2026b. Trisduction: A Linguistically, Topologically, and Mathematically Sealed Verification Architecture. Master Codex, Tractatus Veritatis Trisductivus repository.
- Islam, M. F. 2026c. Operator-independent system-role verification papers. PhilArchive records ISLTFW, ISLTEO-3, ISLTSV.
- Jacobi, C. G. J. 1829. *Fundamenta Nova Theoriae Functionum Ellipticarum*. Königsberg: Borntraeger.
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- Kervaire, M. A. 1958. Non-parallelizability of the  $n$ -sphere for  $n$  greater than 7. *Proceedings of the National Academy of Sciences* 44: 280-283.
- Knus, M.-A., A. Merkurjev, M. Rost, and J.-P. Tignol. 1998. *The Book of Involutions*. Colloquium Publications 44. Providence: American Mathematical Society.
- Landauer, R. 1961. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development* 5: 183-191.
- Mandelstam, L., and I. Tamm. 1945. The uncertainty relation between energy and time in non-relativistic quantum mechanics. *Journal of Physics (USSR)* 9: 249-254.
- Margolus, N., and L. B. Levitin. 1998. The maximum speed of dynamical evolution. *Physica D* 120: 188-195.
- Mayo, D. G. 2018. *Statistical Inference as Severe Testing*. Cambridge: Cambridge University Press.
- Musin, O. R. 2008. The kissing number in four dimensions. *Annals of Mathematics* 168: 1-32.
- Palais, R. S. 1968. The classification of real division algebras. *American Mathematical Monthly* 75: 366-368.
- Popper, K. R. 1959. *The Logic of Scientific Discovery*. London: Hutchinson.
- Schütte, K., and B. L. van der Waerden. 1953. Das Problem der dreizehn Kugeln. *Mathematische Annalen* 125: 325-334.
- Springer, T. A., and F. D. Veldkamp. 2000. *Octonions, Jordan Algebras and Exceptional Groups*. Berlin: Springer.
- Wang, Z., T. Solloway, R. M. Shiffrin, and J. R. Busemeyer. 2014. Context effects produced by question orders reveal quantum nature of human judgments. *Proceedings of the National Academy of Sciences* 111: 9431-9436.
- Weyl, H. 1939. *The Classical Groups: Their Invariants and Representations*. Princeton: Princeton University Press.
- Wilks, S. S. 1932. Certain generalizations in the analysis of variance. *Biometrika* 24: 471-494.
- Zorn, M. 1933. Alternativkörper und quadratische Systeme. *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg* 9: 395-402.

# APPENDIX A • A SELF-CONTAINED PROOF OF THE FROBENIUS CLASSIFICATION

**Theorem (Frobenius 1878).** Every finite-dimensional associative unital division algebra  $A$  over  $\mathbb{R}$  is isomorphic to  $\mathbb{R}$ ,  $\mathbb{C}$ , or  $\mathbb{H}$ .

**Step 1, quadratic minimality.** For  $a \in A$ , the subalgebra  $\mathbb{R}[a]$  is commutative, finite-dimensional, and without zero divisors, hence a field extension of  $\mathbb{R}$ , hence isomorphic to  $\mathbb{R}$  or  $\mathbb{C}$ . So every element satisfies a real polynomial of degree at most two. For  $a \notin \mathbb{R}$  the minimal polynomial is  $x^2 - 2tx + n$  with  $t^2 < n$ , and then  $(a - t)^2 = t^2 - n < 0$ : every element is a real number plus an element whose square is a negative real. The step  $\mathbb{R}[a] \cong \mathbb{R}$  or  $\mathbb{C}$  is the classification of finite field extensions of  $\mathbb{R}$ , equivalent to the fundamental theorem of algebra; it is the one external theorem consumed inside this appendix and is named here as such, the first of the three named consumptions of Section 3.

**Step 2, the trace-zero complement.** Define  $V = \{x \in A : x^2 \in \mathbb{R}, x^2 \leq 0\}$ , with  $0 \in V$ . Then  $V \cap \mathbb{R} = \{0\}$ , and Step 1 gives  $A = \mathbb{R} + V$ . The substantive claim is that  $V$  is a linear subspace. Closure under scalars is immediate. Let  $u, v \in V$  be linearly independent. First,  $1, u, v$  are linearly independent: a relation  $c_0 + c_1u + c_2v = 0$  with  $c_1 \neq 0$  gives  $u = \alpha + \beta v$ , hence  $u^2 = \alpha^2 + \beta^2v^2 + 2\alpha\beta v$ , and  $u^2 \in \mathbb{R}$  with  $v \notin \mathbb{R}$  forces  $\alpha\beta = 0$ ;  $\beta = 0$  puts  $u$  in  $\mathbb{R} \cap V = \{0\}$ , and  $\alpha = 0$  makes  $u, v$  dependent, both contradictions. Next,  $u + v \notin \mathbb{R}$  and  $u - v \notin \mathbb{R}$  by that independence, so Step 1 gives real quadratics  $(u + v)^2 = 2s(u + v) + r$  and  $(u - v)^2 = 2s'(u - v) + r'$ . Adding, the left side is  $2u^2 + 2v^2 \in \mathbb{R}$ , so  $2(s + s')u + 2(s - s')v \in \mathbb{R}$ , and independence of  $1, u, v$  forces  $s + s' = 0$  and  $s - s' = 0$ , hence  $s = s' = 0$  and  $(u \pm v)^2 \in \mathbb{R}$ . For the sign: if  $(u + v)^2 = c \geq 0$ , then  $(u + v - \sqrt{c})(u + v + \sqrt{c}) = 0$ , the division property gives  $u + v = \pm\sqrt{c} \in \mathbb{R}$ , contradicting  $u + v \notin \mathbb{R}$ . So  $(u + v)^2 < 0$  and  $u + v \in V$ . The dependent case  $v = \gamma u$  is covered by scalar closure. Hence  $V$  is a subspace and  $A = \mathbb{R} \oplus V$ .

**Step 3, the inner product.** On  $V$  define  $\langle u, v \rangle = -\frac{1}{2}(uv + vu)$ . The value is real, since  $uv + vu = (u + v)^2 - u^2 - v^2 \in \mathbb{R}$ ; the form is symmetric and bilinear; and  $\langle u, u \rangle = -u^2 > 0$  for  $u \neq 0$ . So  $\langle \cdot, \cdot \rangle$  is a positive-definite inner product, and orthogonality of  $u, v$  means exactly  $uv = -vu$ .

**Step 4, small dimensions.** If  $\dim V = 0$ , then  $A = \mathbb{R}$ . If  $\dim V = 1$ , pick  $u \in V$  with  $\langle u, u \rangle = 1$ : then  $u^2 = -1$  and  $A = \mathbb{R} \oplus \mathbb{R}u \cong \mathbb{C}$ .

**Step 5, dimension two builds the quaternions.** If  $\dim V \geq 2$ , pick orthonormal  $i, j \in V$ :  $i^2 = j^2 = -1$  and  $ij = -ji$ . Set  $k = ij$ . Then  $k^2 = (ij)(ij) = i(ji)j = -i(ij)j = -(i^2)(j^2) = -1$ . Also  $k \notin \mathbb{R}$ : from  $ij = r \in \mathbb{R}$ , left multiplication by  $i$  gives  $-j = ri$ , putting  $j$  in  $\mathbb{R}i$  against independence. So  $k \in V$ . Orthogonality:  $ki + ik = (ij)i + i(ij) = -(ji)i + i^2j = -j(i^2) - j = j - j = 0$ , and  $kj + jk = (ij)j + j(ij) = i(j^2) + (ji)j = -i - (ij)j = -i + i = 0$ . The remaining relations follow:  $jk = j(ij) = (ji)j = -$

$(ij)j = -i(j^2) = i$ , and  $ki = (ij)i = -(ji)i = -j(i^2) = j$ . So  $\{1, i, j, k\}$  spans a subalgebra with  $i^2 = j^2 = k^2 = ijk = -1$ , isomorphic to  $\mathbb{H}$ .

**Step 6, nothing beyond.** Suppose  $w \in V$  is orthogonal to  $i, j$ , and  $k$ , so  $w$  anticommutes with each. Then  $w(ij) = (wi)j = (-iw)j = -i(wj) = -i(-jw) = (ij)w$ , so  $w$  commutes with  $k$ . But orthogonality gives  $wk + kw = 0$ , hence  $2wk = 0$ , hence  $wk = 0$ , and the division property forces  $w = 0$ . Therefore  $V = \text{span}\{i, j, k\}$ ,  $A = \mathbb{R} \oplus V \cong \mathbb{H}$ , and the classification is complete. ■

The exposition follows the classical line of Palais (1968) and Ebbinghaus and colleagues (1991), reproduced here whole so that the completion theorems of Section 4 rest on no external text beyond the Step 1 consumption named above. Every hypothesis consumed, finite dimension, associativity, absence of zero divisors, unital  $\mathbb{R}$ -linearity, is a composition-law clause of Section 4.2.

## APPENDIX B · THE CORROBORATION REGISTER · ZERO LOAD

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Every entry below is a true theorem or exact structure in its own register, every entry corroborates a result of the body from outside algebra, and every entry is removable without motion of any verdict in this paper. The register exists because the architecture's honest-typing law requires corroboration to be named as corroboration, never silently promoted to warrant.

**Euler closure.**  $V - E + F = 2$  at  $V = 4$ ,  $E = 6$ ,  $F = 4$  certifies, from the polyhedral register, the four-slot closure that completeness forces, with directional resolution doubling six edges to twelve transitions.

**Newton-Gregory.**  $K(3) = 12$  (Schütte and van der Waerden 1953): the kissing number of three-dimensional space confirms the twelve from packing, and the face-centered-cubic kissing configuration coincides, at scale  $1/\sqrt{2}$ , with the pure-imaginary norm-2 Hurwitz slice of Section 4.8 by exact set comparison.

**The kissing ladder.**  $K(4) = 24$  (Musin 2008), achieved by the 24-cell: the full second shell whose slice is the twelve, the four-dimensional count sitting exactly one doubling above the three-dimensional one.

**The 24-cell.** The unit Hurwitz shell is the vertex set of the self-dual regular polytope of  $\mathbb{R}^4$ , and the two shells are similar at Euclidean ratio  $\sqrt{2}$  via left multiplication by  $1 + i$ , an element of quadratic norm 2 that carries the twenty-four unit Hurwitz elements bijectively onto the norm-2 shell.

**De Gua.**  $D^2 = A^2 + B^2 + C^2$  on the trirectangular figure: the quadratic face of the verdict identity read in solid geometry, with the Hodge star identifying each leg plane with the axis it omits.

**The slot simplex.** The four frame elements  $\{1, i, j, k\}$  lie pairwise at distance  $\sqrt{2}$  in  $\mathbb{R}^4$ , a regular tetrahedron whose rotation group realizes geometrically the  $A_4$  that Section 4.8 obtains by permutation algebra alone.

**The topological wall.** Poincaré-Brouwer forbids a nonvanishing tangent field on  $S^2$ ; Adams confines H-space spheres to  $S^0$ ,  $S^1$ ,  $S^3$ ,  $S^7$ ; Bott-Milnor and Kervaire confine parallelizable spheres to  $S^1$ ,  $S^3$ ,  $S^7$  and real division structure of any kind to dimensions 1, 2, 4, 8. The body needs only the associative classification, proved in Appendix A; the topology closes the non-associative escape routes from its own register.

**Friedrichs-Hodge.** The three-way orthogonal decomposition of square-integrable differential forms witnesses that the structure type, three mutually orthogonal components closing a function space, is native to analysis. Structural analogy only, load-bearing on nothing, the standing typing of the master codex carried here unchanged.

## APPENDIX C • THE WARRANT LEDGER

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Every load-bearing and supporting claim of this paper, with its warrant tier. The tier travels with the claim in any restatement.

Claim	Warrant
Root Axiom, $\forall x \in \mathbb{U}, \Delta E_k(x) > 0$	Axiom of the architecture; no unconditional mathematical proof claimed
Kinetic floor at the transition register	Theorem grade, conditional on the postulates of quantum dynamics (Mandelstam-Tamm; Margolus-Levitin)
Irreversible-erasure signature	Theorem plus instrument grade for the subclass (Landauer; Bérut)
Static existence under the floor	Premise grade, substrate monism
Triple population of an actuation	Conditional theorem on monism, Omega Boundary register
CL-1 associativity	Premise, typed to the audit-symmetry legislation; under standing empirical exercise
CL-2 integrality	Premise, typed to the Mass Mandate; under standing empirical exercise
CL-3 linearity with ground identity	Premise, typed to the pipeline's linear register; under standing empirical exercise
Floor 2 order-sensitivity	Premise, typed to operational measurement asymmetry P3
Order-Sensitivity Completion	Theorem, conditional on CL-1, CL-2, CL-3, via Appendix A
Finite field extensions of $\mathbb{R}$ (FTA), Appendix A Step 1	External consumption, theorem grade, named
Triaxial Forcing, $A \cong \mathbb{H}$ , exactly three axes	Theorem, conditional on CL-1, CL-2, CL-3 and order-sensitivity
Floor 1 axis plurality	Corollary at this register
No fourth axis	Corollary of the same classification
Conjugation unique with fixed subspace exactly $\mathbb{R}$	Theorem, proof in text
$Z(\mathbb{H}) = \mathbb{R}$	Theorem, proof in text
Ground named the registration line	Definitional identification, premise-typed

Claim	Warrant
Twelve, W1 slot combinatorics	Theorem, conditional on the forced carrier and Floor 2
Twelve, W2 the $A_4$ torsor	Theorem, proof in text
The 2T double cover and class equation	Classical identification (Conway-Smith) plus exhaustive enumeration
Twelve, W3 shell arithmetic $24 = 12 + 12$	Theorem, proof in text; Jacobi corroboration
Gate content	Operational import, Operational Content Theorem; never derived from algebra
Verdict identity $\det(R) = \lambda^2$	Theorem, proof in text
Pipeline factorization	Theorem, proof in text
Bounds $\det(R), \det(G) \in [0, 1]$	Theorem, inline: norm multiplicativity (§4.4) and AM-GM on the trace-3 correlation matrix; Hurwitz and Hadamard retired from warrant
Catalog completeness	Theorem, via Weyl's first fundamental theorem
Frame invariance $\operatorname{Re}(rw\bar{r}) = \operatorname{Re}(w)$	Theorem, proof in text
Return Law, $\text{GOL} \iff \lambda \neq 0$	Theorem at this register, closed form of APEX-PSP-MU-01
Precedence law and the collapse floor $\varepsilon$	Legislation plus numerical-analysis engineering
Quantization mapping $Q$	Validated engineering, outside this paper's claims
Recorded battery, M-CHK. 1 through M-CHK.16	Engineering grade, re-runnable, falsifier-bearing
Appendix B register	Corroboration, zero load, removable without verdict motion

## VERDICT ON THIS PAPER

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**[v] The mathematics-alone seal stands.** Theorem grade on every stated computation and on the named external roster, with the one long classification proved whole at Appendix A. Conditional theorem grade on the completion chain, premises CL-1, CL-2, CL-3 and order-sensitivity, typed to their legislative sources and standing under empirical exercise. Premise grade where monism is carried. Operational import, fenced and named, on gate content. Corroboration grade, zero load, on the whole of Appendix B. Engineering grade on the sixteen-check recorded battery, residues at machine precision, integer core exact. The independence law is satisfied at the third face: the linguistic and geometric instrument sets are removed entirely, and no verdict moves. One premise, order-sensitivity, carries the triaxial count and the directed roster together. The word counted three; the geometry closed four; the algebra is the receipt.

Provenance. Source: the Trisduction Master Codex, sealed 11 June 2026 (Islam 2026b), Part II read at its mathematical layer. This standalone forged 11 June 2026; battery executed at forge time, IEEE-754 double precision, numpy 2.4.4, seed 20260611. Substrate-portable per PSP-005: any sufficiently capable substrate reproduces every verdict in this document from the statements as printed.